MASHUK ELAHI

Marine & Offshore Electrical Engineer | Hybrid Power, BESS & Commissioning

PERSONAL INFORMATION

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PROFESSIONAL SUMMARY

Electrical engineer with 16 years of experience in marine, offshore and vessel newbuild work: LV/MV distribution, single-line diagrams, load lists and cable schedules, MV 11 kV switchgear, protection systems, battery energy storage (BESS) and hybrid power. I take systems from Factory Acceptance Testing through class approval and energisation to handover, and I build Python tools that keep engineering documents consistent. Earlier work includes semiconductor critical-power (30 MW / 33 MVA, zero audit rework), cogeneration and IECEx/ATEX hazardous areas. CDCP certificate held; ISO 9001 and ISO 50001 Lead Auditor. Open to roles across Singapore, Europe and the UAE.

CORE COMPETENCIES

- **Power systems:** MV/LV distribution; MV 11 kV switchgear; transformers, protection relays, and grounding; UPS, standby generators, and BESS integration
- **Controls & automation:** EPMS/SCADA, BMS/BAS, and PLC/SCADA; ABB 800xA; IEC 61850; engineering workflow automation with Python and AutoCAD Electrical
- **Commissioning & reliability:** FAT/SAT, point-to-point checks, energisation, functional performance and integrated systems testing; N+1, 2N/2N+1 redundancy; Uptime Institute Tier topologies; 99.99%+ uptime SLAs and concurrent maintainability; MOP/SOP/EOP; critical cooling interfaces (chiller plant, CRAH/CRAC, PCW)
- **Energy, hazardous areas & risk:** PUE tracking and energy-performance monitoring (ISO 50001); cogeneration and autonomous renewables; IECEx/ATEX design; marine electrical practice to IEC 60092-201; compliance auditing and gap analysis; technical bid evaluation and FMEA

WORK EXPERIENCE

Electrical Engineer | Hybrid & Critical-Power Systems, Strategic Marine Pte. Ltd., Singapore
Jul 2024 to Present

- Designed 450 ekW three-generator LV distribution with SLDs, load lists, and cable schedules; first-pass approval from four certification bodies with no rework.
- Engineered aluminium-hull IT earthing architecture with single-point common-zero bonding to IEC 60092-201, and resolved stray-current issues on fast crew boats by upgrading isolation switches to DPST.
- Integrated a 72 kWh lithium-ion BESS and BMS into a hybrid power system that cut fuel draw by up to 50% at low load; ran the BMS-to-EPMS interface and FAT through to handover.
- Tracked electrical-system weights exceeding 7,800 kg across load lists and cable schedules, feeding vessel stability calculations.
- Automated SLD generation and VDR alarm-list production using Python, AI agents, and AutoCAD Electrical scripting.

Senior Project Engineer | Semiconductor Critical-Power Commissioning, Exyte, Singapore
Feb 2023 to Jul 2024

- Ran 300 mm semiconductor cleanroom critical-power hook-up and energisation on 30 MW (33 MVA) of critical load at 1.5 kW/m² fab power density, with zero audit rework and first-pass PLC/SCADA interlock validation against 100% tool-uptime and N+1 objectives.
- Introduced prefabricated power skids that cut on-site installation labour by 50%, coordinating with chilled-water and process-cooling-water (PCW) commissioning teams.

E&I Engineer | Industrial Power, Controls & Hazardous Areas, Phoenix Mecano S.E.A. Pte. Ltd., Singapore
Jan 2021 to Jan 2023

- Owned the E&I scope for a 10 MW cogeneration plant (transformers, MV/LV switchgear, 400 kVA UPS, SCADA integration, tie-in design, instrument loops) and led technical bid evaluation.
- Designed IECEx/ATEX hazardous-area systems, including Ex equipment selection, intrinsically safe barriers, and zoning studies.
- Delivered Malaysia's first solar-wind autonomous remote-power installation on a 24 V DC solar-battery bus.

E&I Superintendent | Offshore Power Systems & Commissioning, Keppel FELS / Keppel Offshore & Marine, Singapore
Nov 2012 to Dec 2020

- Led MV 11 kV and ABB 800xA commissioning with crews of up to 100 people.
- Energised an offshore-wind substation to full IEC 61850 conformity with zero punch-list items.
- Authored 30 E&I risk assessments and mentored 15 engineers through Innovation Quality Circles.

Assistant Electrical Engineer | Vessel Newbuild Electrical, Ananda Shipyard & Slipways Ltd., Bangladesh
Sep 2009 to Oct 2012

- Supported installation, testing, and commissioning on vessel newbuilds.

EDUCATION AND TRAINING

- Master's Degree in Continuing Education in Sustainable Electricity Generation (UCAM, 60 ECTS, título propio), UCAM Universidad Católica de Murcia, Spain, Apr 2026
- MSc Digital Transformation, London Metropolitan University, UK, Mar 2026
- Master of Business Administration (MBA), University of Sunderland, UK, Jun 2025
- BSc Electrical and Electronic Engineering, Bangladesh University, Jul 2008
- Diploma in Electronics Engineering, Jessore Polytechnic Institute, Jan 2004

CERTIFICATIONS AND TRAINING COURSES

- Certified Data Centre Professional (CDCP), EPI, 2026
- ISO 50001:2018 Energy Management Lead Auditor, GAQM / IGC
- ISO 9001:2015 Quality Management Lead Auditor, GAQM / IGC
- IECEx Ex 001: Basic Principles of Protection in Explosive Atmospheres, valid through Oct 2027
- ABB Power System / LV Switchboard Training
- Fluke DSX2-8000 Cable Analyzer / Copper Certifier Training

LANGUAGE SKILLS

English

DIGITAL SKILLS

ABB 800xA; PLC/SCADA systems; IEC 61850 protection relays; EPMS/SCADA; AutoCAD Electrical; Python (engineering workflow automation); Fluke DSX2-8000 Cable Analyzer / Copper Certifier.